Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/10/2016
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Research Opportunities in Combustion Science		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	02/01/2010	**End Date:**	01/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:	ISS (for BASS) NOTE: End date is now 1/31/2016 per NSSC information (Ed., 1/14/2015) NOTE: End date is now 1/31/2015 per NSSC information (Ed., 3/12/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ferkul, Paul (Universities Space Research Association) Olson, Sandra (NASA Glenn Research Center)		
Grant/Contract No.:	NNX10AD99A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, shoulder, and wake regions results in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with several imbedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples. In addition to James T'ien (PI), Dr. Paul Ferkul (NCSER) and Dr. Sandra Olson (NASA GRC) will participate as Co-Investigators in the project. Ed. note (June 2016): Project continued by "Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)," grant NNX16AE84A, with the same Principal Investigator.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	1. Numerical modeling for BASS experiments 1.1 Flat SIBAL samples Flame growth, spread, and quenching extinction over a thin composite cotton-fiberglass fabric blend (referred to as the SIBAL fabric) were studied in low-speed concurrent purely forced flows aboard the International Space Station. The tests were conducted in a small flow duct within the Microgravity Science Glovebox. The fuel samples measured 1.2 and 2.2 cm wide and 10 cm long. Ambient oxygen was varied from 21% down to 16% molar concentration and flow speed from 55 cm/s down to 1 cm/s. A slow purely forced flow resulted in a small flame, enabling us to observe the entire history of flame development including ignition, flame growth, steady spread (in some cases), and decay at the end of the sample. In addition, by decreasing flow velocity during some of the tests, low-speed flame quenching extinction limits were determined as a function of oxygen percentage. The quenching speeds were found to be between 1 and 5 cm/s with higher extinction speeds in lower oxygen atmospheres. The shape of the quenching boundary supports the prediction by earlier theoretical models. These long duration microgravity experiments provide a rare opportunity for solid fuel combustion since microgravity time in ground-based facilities is generally not sufficient. This is the first time that a low-speed quenching boundary in concurrent spread is mapped in a clean and unambiguous manner. A previously developed three-dimensional transient model is modified to compare with the experiment. The modification includes the use of two-step SIBAL fabric pyrolysis kinetics where the rate constants are determined using Thermo-Gravimetric Analysis data. The model yields good quantitative comparison on the quenching boundary, the flame transient development, and the steady flame spread rates (References #1 and #2). 1.2 Radiation Model and Spherical PMMA samples Gas phase radiative heat transfer becomes especially important for flame spread over solid fuel in microgravity near the low velocity quenching limit, also known as radiative quenching limit. Near this limit the flame color of PMMA is blue due to the low soot content and therefore, spectral dependency of the radiative heat transfer becomes important. Commonly used radiation treatments such as gray gas model and wide band model face difficulty in obtaining accurate result in such condition. Moreover, being a diffusion flame prevents the usage of some of the popular special approximations such as spherical harmonics method since the absorption coefficient becomes zero in some wave lengths. This research overcomes such challenges by solving the radiation field around a PMMA sphere by Statistical Narrow-Band Correlated-k (SNB-CK) spectral model combined with three dimensional discrete ordinate method. The model is applied to an experimental configuration from the recent microgravity BASS project aboard the International Space Station. A PMMA sphere (initial diameter of 2 cm) is placed in a small wind tunnel (8 cm x 8 cm x 17.5 cm) within the microgravity glovebox where flow speed and oxygen percentage are varied. The recorded flame spread rate after ignition at the forward stagnation point and radiative quenching limiting flame when flow velocity is reduced are compared with the model results. (PhD dissertation by Makoto Endo and Reference #3) References: 1. Zhao, X. and T'ien, J.S., A Three-dimensional Transient Model for Flame Growth and Extinction in Concurrent Flows, Combustion and Flame 162 (2015) 1829-1839, http://dx.doi.org/

	2. Zhao, X., Liao, Y.-T. T., Johnston, M.C., T'ien, J.S., Ferkul, P.V. and Olson, S.L., Concurrent flame growth, spread and quenching over composite fabric samples in low speed purely forced flow in microgravity, to be presented at the Thirty-Six International Symposium on Combustion, July 31-August 5, 2016, Seoul, Korea. 3.Endo, Makoto, "Numerical modeling of flame spread over spherical solid fuel under low speed flow in microgravity: model development and comparison to space flight experiments", PhD dissertation, Case Western Reserve University, May, 2016.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Zhao X, T'ien JS. "A three-dimensional transient model for flame growth and extinction in concurrent flows." Combustion and Flame. 2015 May;162(5):1829-39. http://dx.doi.org/10.1016/j.combustflame.2014.12.003 , May-2015
Articles in Peer-reviewed Journals	Jomaas G, Torero JL, Eigenbrod C, Niehaus J, Olson SL, Ferkul PV, Legros G, Fernandez-Pello AC, Cowlard AJ, Rouvreau S, Smirnov N, Fujita O, T'ien JS, Ruff GA, Urban DL. "Fire safety in space – beyond flammability testing of small samples." Acta Astronautica. 2015 Apr-May;109:208-16. http://dx.doi.org/10.1016/j.actaastro.2014.11.025 , May-2015
Awards	T'ien JS. (James S. T'ien) "AIAA Space Processing Award, July 2015." Jul-2015
Dissertations and Theses	Endo M. (Makoto Endo) "Numerical modeling of flame spread over spherical solid fuel under low speed flow in microgravity: model development and comparison to space flight experiments." PhD dissertation, Case Western Reserve University, May 2016. , May-2016
Papers from Meeting Proceedings	Zhao X, Tien JS, Ferkul PV, Olson SL. "Concurrent flame growth, spread, and quenching over composite fabric samples in low speed purely forced flow in microgravity." 9th US National Combustion Meeting, Cincinnati, Ohio, May 17-20, 2015. 9th US National Combustion Meeting, Cincinnati, Ohio, May 17-20, 2015. , May-2015